Introduction to Machine Learning

Lecture 5: Principal Component Analysis - Part II

Ali Bereyhi

ali.bereyhi@utoronto.ca

Department of Electrical and Computer Engineering

University of Toronto

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Latent Space $\leftrightsquigarrow$ Reconstruction

Dimensionality Reduction: *Latent Variable*

Dimensionality is reduced linearly via $\mathbf{U}^{\mathsf{T}} : \mathbb{R}^D \mapsto \mathbb{R}^K$ *as*

$$\boldsymbol{z}_n = \mathbf{U}^{\mathsf{T}} \left(\boldsymbol{x}_n - \boldsymbol{\mu} \right)$$

We call $\boldsymbol{z}_n$ *latent variable*

Higher Dimensional Recovery: *Reconstruction*

We reconstruct a sample $\boldsymbol{x}_n \in \mathbb{R}^D$ *from its latent variable* $\boldsymbol{z}_n \in \mathbb{R}^K$ *as*

$$\hat{\boldsymbol{x}}_n = \mathbf{U} \boldsymbol{z}_n + \boldsymbol{\mu}$$

Dimensionality Reduction as *Learning Task*

- Data
  - ↳ *Collection of samples* $\mathbb{D} = \{\boldsymbol{x}_n : n = 1, \ldots, N\}$
- Model
  - ↳ *Compression* $\boldsymbol{z} = \mathbf{U}^{\mathsf{T}} (\boldsymbol{x} - \boldsymbol{\mu})$
  - ↳ *Decompression* $\hat{\boldsymbol{x}} = \mathbf{U}\boldsymbol{z} + \boldsymbol{\mu}$
- Learning algorithm
  - ↳ *We look for algorithm* $\mathcal{A} : \mathbb{D} \mapsto \mathbf{U}^\star, \boldsymbol{\mu}^\star$
  - ↳ $\mathbf{U}^\star, \boldsymbol{\mu}^\star$ *are good ones!*

Learning via *Minimum Representation Error*

Optimal $\mathbf{U}^\star$ *and* $\boldsymbol{\mu}^\star$ *are defined as*

$$\mathbf{U}^\star, \boldsymbol{\mu}^\star = \underset{\mathbf{U}, \boldsymbol{\mu}}{\operatorname{argmin}} \mathcal{J}(\mathbf{U}, \boldsymbol{\mu}) = \underset{\mathbf{U}, \boldsymbol{\mu}}{\operatorname{argmin}} \frac{1}{N} \sum_{n=1}^{N} \|\hat{\boldsymbol{x}}_n - \boldsymbol{x}_n\|^2$$

Alternative Formulation

Optimal bias is the *dataset sample mean* $\equiv$ *centroid*

$$\boldsymbol{\mu}^\star = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n$$

For design of $\mathbf{U}$, we can use an alternative formulation

Dimensionality Reduction via *Maximal Representation Variance*

Optimal $\mathbf{U}^\star$ that minimizes recovery error equivalently *maximizes variance*

$$\mathbf{U}^\star = \underset{\mathbf{U}}{\operatorname{argmax}} \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{z}_n\|^2$$

Notion of Correlation in Data

? *Why do we call it variance?*

We know that $\mathbf{U}^\mathsf{T}(\boldsymbol{x}_n - \boldsymbol{\mu}^\star) = \boldsymbol{z}_n$, so can say

$$\|\boldsymbol{z}_n\|^2 = \|\mathbf{U}^\mathsf{T}\boldsymbol{x}_n - \mathbf{U}^\mathsf{T}\boldsymbol{\mu}^\star\|^2 = \|\mathbf{U}^\mathsf{T}\boldsymbol{x}_n - \frac{1}{N}\sum_{n=1}^{N}\mathbf{U}^\mathsf{T}\boldsymbol{x}_n\|^2$$
$$= \|\mathbf{U}^\mathsf{T}\boldsymbol{x}_n - \mathtt{avg}\left(\mathbf{U}^\mathsf{T}\mathbb{D}\right)\|^2$$

Therefore, we could say

$$\frac{1}{N}\sum_{n=1}^{N}\|\boldsymbol{z}_n\|^2 = \mathtt{var}\left(\mathbf{U}^\mathsf{T}\mathbb{D}\right)$$
$$= \textit{variance in the latent space}$$

Notion of Variance in Data

- ✔ Let's start with $K = 1$

Recall that $z_n = \mathbf{u}^\mathsf{T} \left(\boldsymbol{x}_n - \boldsymbol{\mu}^\star \right)$ and let

$$\tilde{\boldsymbol{x}}_n = \boldsymbol{x}_n - \boldsymbol{\mu}^\star \frac{1}{N} \sum_{n=1}^{N} \boldsymbol{x}_n$$

So, we have $z_n = \mathbf{u}^\mathsf{T} \tilde{\boldsymbol{x}}_n$; thus,

$$\left(z_n \right)^2 = \mathbf{u}^\mathsf{T} \tilde{\boldsymbol{x}}_n \tilde{\boldsymbol{x}}_n^\mathsf{T} \mathbf{u}$$

Notion of Variance in Data

We could hence compute the variance as

$$\begin{aligned}\frac{1}{N}\sum_{n=1}^{N}|z_n|^2 &= \frac{1}{N}\sum_{n=1}^{N}\mathbf{u}^\mathsf{T}\tilde{\boldsymbol{x}}_n\tilde{\boldsymbol{x}}_n^\mathsf{T}\mathbf{u}\\ &= \mathbf{u}^\mathsf{T}\left(\frac{1}{N}\sum_{n=1}^{N}\tilde{\boldsymbol{x}}_n\tilde{\boldsymbol{x}}_n^\mathsf{T}\right)\mathbf{u}\\ &= \mathbf{u}^\mathsf{T}\left(\frac{1}{N}\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\mathsf{T}\right)\mathbf{u}\end{aligned}$$

where we define

$$\tilde{\mathbf{X}} = [\tilde{\boldsymbol{x}}_1,\dots,\tilde{\boldsymbol{x}}_N] \in \mathbb{R}^{D\times N}$$

Sample Covariance

Sample Covariance

The sample covariance of the dataset is given by

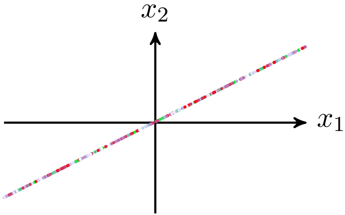
$$\mathbf{\Sigma} = \frac{1}{N} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^{\mathsf{T}}$$

In case of $K = 1$, we can write the variance as

$$\begin{aligned} \frac{1}{N} \sum_{n=1}^{N} |z_n|^2 &= \mathbf{u}^{\mathsf{T}} \left(\frac{1}{N} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^{\mathsf{T}} \right) \mathbf{u} \\ &= \mathbf{u}^{\mathsf{T}} \mathbf{\Sigma} \mathbf{u} \end{aligned}$$

Sample Covariance

Let's compute sample covariance for the line example

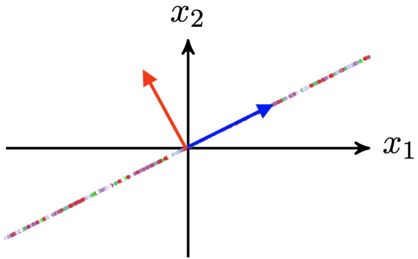


$$\mathbf{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_5] = \begin{bmatrix} -2 & -1 & 0 & 1 & 2 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix}$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 20 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

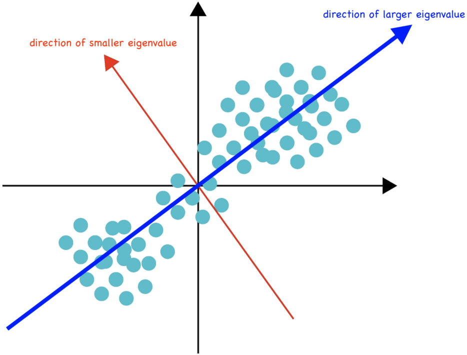
Sample Covariance

The large eigenvalue shows us where the data spans more



Sample Covariance

This applies to any dataset



! *It can tell us right direction for dimensionality reduction!*

Optimal Transform $\equiv$ *Principal Components of Covariance*

$\boldsymbol{\Sigma}$ is positive semi-definite: *we thus have*

$$\boldsymbol{\Sigma} = \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_D \end{bmatrix} \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_D \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^{\mathsf{T}} \\ \vdots \\ \mathbf{v}_D^{\mathsf{T}} \end{bmatrix}$$

with $\lambda_1, \ldots, \lambda_D \geqslant 0$. *Let's replace in the variance expression*

$$\begin{aligned} \frac{1}{N} \sum_{n=1}^{N} |z_n|^2 &= \mathbf{u}^{\mathsf{T}} \boldsymbol{\Sigma} \mathbf{u} \\ &= \sum_{i=1}^{D} \lambda_i \left(\mathbf{u}^{\mathsf{T}} \mathbf{v}_i \right)^2 \end{aligned}$$

Optimal Transform $\equiv$ *Principal Components of Covariance*

We want to find optimal $\mathbf{u}$: *it should maximize the following term*

$$\frac{1}{N}\sum_{n=1}^{N}|z_n|^2 = \sum_{i=1}^{D}\lambda_i\left(\mathbf{u}^\mathsf{T}\mathbf{v}_i\right)^2$$

- $\mathbf{v}_i$*'s are bases; thus, we always have*

$$\left(\mathbf{u}^\mathsf{T}\mathbf{v}_i\right)^2 \leqslant 1$$

- $[\mathbf{v}_1 \cdots \mathbf{v}_D]$ *is an orthonormal matrix; thus,*

$$\|\mathbf{u}^\mathsf{T}[\mathbf{v}_1 \cdots \mathbf{v}_D]\| = \sum_{i=1}^{D}\left(\mathbf{u}^\mathsf{T}\mathbf{v}_i\right)^2 = 1$$

Optimal Transform $\equiv$ *Principal Components of Covariance*

We want to find optimal $\mathbf{u}$: *its should be set to*

$$\mathbf{u} = \mathbf{v}_{\max}$$

where $\mathbf{v}_{\max}$ *corresponds to the maximum eigenvalue* $\lambda_{\max}$

$$\frac{1}{N} \sum_{n=1}^{N} |z_n|^2 = \lambda_{\max}$$

Optimal Transform $\equiv$ *Principal Components of Covariance*

Say $K = 2$, i.e., $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2] \rightsquigarrow z_{n,1}, z_{n,2}$

$$\frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{z}_n\|^2 = \frac{1}{N}\sum_{n=1}^{N} |z_{n,1}|^2 + \frac{1}{N}\sum_{n=1}^{N} |z_{n,2}|^2$$
$$= \mathbf{u}_1^\top \boldsymbol{\Sigma} \mathbf{u}_1 + \mathbf{u}_2^\top \boldsymbol{\Sigma} \mathbf{u}_2$$

With same lines of derivations, we could say

- *Optimal $\mathbf{u}_1$ is $\mathbf{v}_{\max_1}$ corresponding to the largest eigenvalue*
- *Optimal $\mathbf{u}_2$ is $\mathbf{v}_{\max_2}$ corresponding to the second largest eigenvalue*

Principal Component Analysis: *General Algorithm*

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PCA():
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1: Set $\boldsymbol{\mu}$ to the centroid of dataset

$$\boldsymbol{\mu} = \frac{1}{N} \sum_{n=1}^{N} \boldsymbol{x}_n$$

2: Construct the sample covariance matrix

$$\boldsymbol{\Sigma} = \frac{1}{N} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^{\mathsf{T}}$$

3: Decompose $\boldsymbol{\Sigma}$ in terms of its eigenvalues an eigenvectors as $\boldsymbol{\Sigma} = \mathbf{V} \boldsymbol{\Lambda} \mathbf{V}^{\mathsf{T}}$

4: Find the K largest eigenvalues and set $\mathbf{u}_k = \mathbf{v}_{\max_k}$

5: Set the projection matrix to

$$\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_K]$$

Formulating the Problem

We can look at an image as a data sample

This is a 300×200 *image*

we can look at it as a 60000*-dimensional sample* $\boldsymbol{x}_n$

Formulating the Problem

? *How many samples then we need?*

! *Obviously large enough to build sample covariance* $\boldsymbol{\Sigma}$

Recall that the sample covariance is of the size $\boldsymbol{\Sigma} \in \mathbb{R}^{D \times D}$

? *This would be a huge matrix to decompose!*

We can alternatively look at the image as

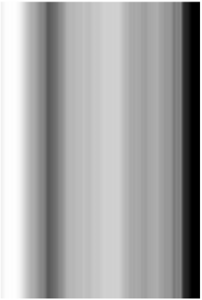
$\mathbf{X} \in \mathbb{R}^{300 \times 200}$ *and treat each column as a single* 300*-dimensional sample*

So, the image itself is a dataset!

$$\mathbf{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_{200}]$$

Using PCA to Compress Images

- ? *How well we can compress?*
- ! *Let's look at the case with* $K = 0$*!*



Using PCA to Compress Images

? *How well we can compress?*

! *Let's look at the case with* $K = 10$*!*

Using PCA to Compress Images

- ? *How well we can compress?*
- ! *Let's look at the case with $K = 20$!*

Using PCA to Compress Images

? *How well we can compress?*

! *Let's look at the case with $K = 30$!*

Using PCA to Compress Images

- ? *How well we can compress?*
- ! *Let's look at the case with $K = 40$!*

Using PCA to Compress Images

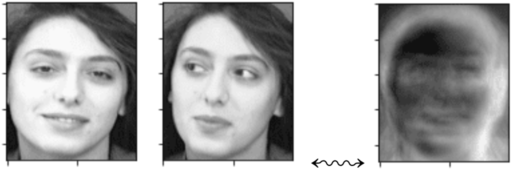
- ? *How well we can compress?*
- ! *Let's look at the case with* $K = 50$*!*

Sounds to be good enough!

Sample Example: *Face Recognition*

One classic application of PCA is face recognition

- *We compress high-dimensional images via PCA to latent space*
- *We use the latent representations to compare two pictures*
  ↳ *If they belong to the same person, latent representations should be close*
- *If we choose K properly then we could have right latent representation*



Formulating the Problem

? *How to find missing rating?*

Formulating the Problem as PCA

? *How to find missing rating?*

We can look at it as a completion problem

$$\text{rating user } n \ = \boldsymbol{x}_n = \begin{bmatrix} \text{movie } 1 \\ \vdots \\ \text{movie } D \end{bmatrix} = \sum_{k=1}^{K} z_{n,k} \mathbf{u}_k$$

- *There are K eigenratings!*
- *Each user is almost perfectly by its latent representation $\boldsymbol{z}_n$*

It sounds like PCA!

Sample Example: *Recommendation System*

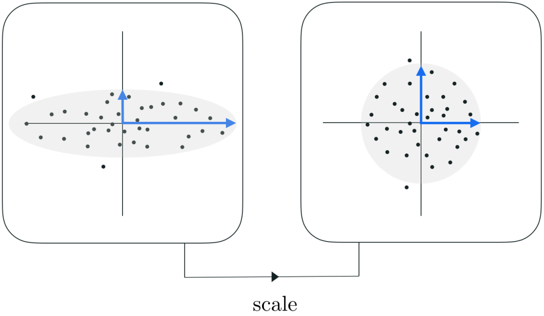
This approach is used in many recommendation systems

- *Our ratings in online shops are used to find the eigenratings*
- *Eigenratings approximate our true ratings from latent space*
- *The shop will the decide what to advertise*

In 2009, this idea won the $ 1M Netflix Prize

Sensitivity to Scaling

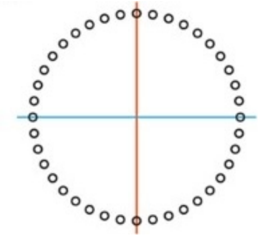
! *PCA is sensitive to scaling*



! *We typically normalize the data before applying PCA*

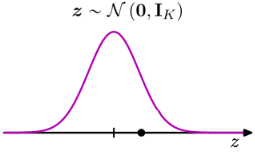
PCA is Linear!

! *PCA cannot capture nonlinear patterns*



! *Extending PCA to nonlinear patterns gave birth to Autoencoders*

PCA as Maximum Likelihood Estimation



We can find $\boldsymbol{\mu}$ and $\mathbf{U}$ by Gaussian maximum likelihood

↳ *This ends up with the same result*

↳ *What if we consider another model for $\boldsymbol{x}$?!* $\rightsquigarrow$ *Autoencoders do*

Further Read

- Bishop
  - ↳ Chapter 12: *Section 12.1* — ***PCA***
- ESL
  - ↳ Chapter 14: *Section 14.1* — ***Principal Components***
- Goodfellow
  - ↳ Chapter 2 — ***Review on Linear Algebra***
  - ↳ Chapter 5: *Section 5.8.1* — ***PCA***
- MacKay
  - ↳ Chapter 34 — ***Latent Space Design***